

Notes: Chen, Chen, Kang, and Peng (JFE, 2020)

Board structure, director expertise, and advisory role of outside directors

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1 Executive overview

1.1 The question

The paper asks how an *exogenous shock to firms' demand for a specific type of expertise* changes:

- (i) board composition (board structure in the “expertise” dimension),
- (ii) the advisory role of outside directors in a setting where expertise should matter (investments involving China),
- (iii) market and labor-market valuation of expertise (appointment announcement returns and subsequent board seats).

1.2 The shock and the central identification idea

The shock is the U.S. Congress granting *Permanent Normal Trade Relations (PNTR)* status to China in October 2000. The key economic change is that PNTR removed the risk of *sudden tariff spikes* on imports from China, which increased the expected profitability of China-related production/trade links and (importantly) changed incentives for U.S. firms to interact with Chinese counterparties.

The empirical strategy uses:

- cross-industry variation in exposure to PNTR, measured by the *NTR gap* (the difference between the high “non-NTR” tariff rate and the normal NTR rate),
- time variation (pre vs post PNTR),
- firm fixed effects to absorb time-invariant heterogeneity in board structure,
- year fixed effects to absorb aggregate trends in director supply, globalization, and corporate governance.

1.3 The main findings

Firms in industries more exposed to PNTR (higher NTR gaps) increase the fraction of outside directors with China-related experience after 2000. This result is robust to alternative measures (e.g., China import penetration) and to instrumental variables designs. Directors with China experience appear to improve the board’s *advisory* role in China-related corporate investments: firms with more of these directors have higher announcement returns around China deals and better post-deal operating performance. The stock market reacts more positively when firms appoint a director with China experience, and the director labor market rewards these directors with additional board seats after PNTR. Finally, the value-enhancing effect is concentrated among directors with China experience who are domiciled in the U.S., while directors domiciled abroad are associated with weaker meeting attendance and (in one specification) more restatements.

2 Institutional background and hypotheses

2.1 Why PNTR changes the demand for directors with China experience

Cross-border investments and partnerships in China require:

- understanding legal and political risks,
- navigating business culture and negotiation norms,
- building networks with local partners and suppliers,
- anticipating regulatory frictions and compliance constraints.

PNTR makes China-related activities more attractive and less uncertain, which increases firms' incentives to acquire *China-specific expertise* on the board.

2.2 Hypotheses

Hypothesis 1 (board composition): The proportion of outside directors with China-related experience increases after PNTR, especially for firms in industries with larger exposure to PNTR.

Hypothesis 2 (advisory role): Firms with more outside directors with China experience realize higher announcement returns and better post-deal operating performance for China-related investments (M&As, joint ventures, strategic alliances involving Chinese counterparties).

3 Data and measurement

3.1 Sample

The main samples are:

- **Firm-year sample:** U.S. manufacturing firms, 1996–2011, with board data (BoardEx, RiskMetrics) and financial/return data (Compustat/CRSP), resulting in about 14k firm-year observations.
- **Director-year sample:** outside director-year observations (BoardEx/RiskMetrics), used to describe director attributes and for labor-market analyses.
- **China deal sample:** 234 completed China deals (98 M&As, 56 JVs, 80 SAs) from SDC, matched to board/return data.

3.2 Key variables

Director with China experience. An outside director is coded as having China experience if her prior career includes work in China or a China-focused division/subsidiary of a firm (in roles such as director, CEO/CFO/COO, chairman/president/VP/manager, regional executive roles, etc.).

Board composition measure.

$$China-director_{i,t} = \frac{\#\{\text{outside directors with China experience}\}}{\#\{\text{all directors on board}\}} \in [0, 1]. \quad (1)$$

Industry exposure to PNTR: NTR gap. For industry j , the *NTR gap* is:

$$NTR\ gap_{j,1999} \equiv non-NTR\ tariff_{j,1999} - NTR\ tariff_{j,1999}.$$

Intuition: industries with larger gaps face larger “policy uncertainty wedges” pre-PNTR and thus experience a larger reduction in uncertainty once PNTR is granted.

Post indicator.

$$Post_t = \mathbf{1}\{t \geq 2000\}.$$

(The paper uses 2000–2011 as the post period.)

Alternative exposure: China import penetration (CIP). CIP is the ratio of imports from China to domestic absorption (shipments + imports – exports), and is used both as an alternative exposure measure and as a robustness check with an IV based on non-U.S. countries.

4 Model 1: Difference-in-differences for board composition

4.1 Baseline DiD specification (Equation (1) in the paper)

Let i index firms, j industries, and t years. The paper estimates:

$$China-director_{ijt} = \beta_0 + \beta_1 (NTR\ gap_{j,1999} \times Post_t) + \beta_2' X_{ijt} + \delta_t + \theta_i + \varepsilon_{ijt}, \quad (2)$$

where:

- X_{ijt} includes board and firm controls commonly used in the board-structure literature (e.g., log board size, independence, firm size, Tobin’s q , volatility, diversification, leverage, profitability, foreign sales).
- δ_t are year fixed effects.
- θ_i are firm fixed effects.

4.2 Identification logic

β_1 compares:

- (i) the change in *China-director* after 2000 vs before 2000 (*time difference*), and
- (ii) how that change differs between high- and low-exposure industries (*cross-sectional difference*).

The key assumption is a **parallel-trends** type statement: absent PNTR, high- and low-gap industries would have had similar trends in appointing China-experience directors, after controlling for firm and year fixed effects and observable covariates.

4.3 How to interpret magnitudes

Because *China-director* is a fraction, a coefficient of 0.012 on $NTR\ gap \times Post$ means:

- a 0.10 increase in NTR gap (10 percentage points in the tariff gap) raises *China-director* by about 0.0012 (0.12 percentage points) after 2000, relative to before.
- a move from the 25th to the 75th percentile of NTR gap (roughly 0.228 to 0.400) raises *China-director* by about $0.012 \times (0.172) \approx 0.0021$, i.e., about 0.21 percentage points.

Given that pre-PNTR average *China-director* is economically small, this is a large proportional change.

5 Model 2: Instrumental variables for board composition

5.1 Why IV?

Even with firm and year fixed effects, one might worry that unobserved industry characteristics correlated with NTR gaps also affect the post-2000 demand for China-experience directors. The paper therefore instruments for exposure.

5.2 IV using Smoot–Hawley non-NTR rates

The instrument is the Smoot–Hawley-based non-NTR tariff rate in 1990, interacted with *Post*. The 2SLS structure is:

$$\begin{aligned} \textbf{First stage: } (NTR\ gap_{j,1999} \times Post_t) &= \pi_0 + \pi_1 (SmootHawley\ non-NTR_{j,1990} \times Post_t) \\ &\quad + \pi_2' X_{ijt} + \delta_t + \theta_i + u_{ijt}, \end{aligned} \tag{3}$$

$$\textbf{Second stage: } China-director_{ijt} = \beta_0 + \beta_1 \widehat{(NTR\ gap_{j,1999} \times Post_t)} + \beta_2' X_{ijt} + \delta_t + \theta_i + \varepsilon_{ijt}. \tag{4}$$

Economic intuition for exclusion: a tariff schedule set decades earlier should affect post-2000 director demand only through its predictive power for the NTR gap.

5.3 IV using non-U.S. China import penetration

For the alternative exposure measure (CIP in the U.S.), the paper instruments with China import penetration in eight other high-income countries. The logic is that common global supply shocks to China exports affect many rich countries, generating strong first-stage relevance, while the non-U.S. penetration should not directly change U.S. firms' director demand except through U.S. penetration.

6 Advisory role test: China deals and firm value

6.1 Event-study construction of abnormal returns

For a deal announced at date 0, estimate a market model on an estimation window (210 trading days ending 11 days before the announcement):

$$r_{i,\tau} = \alpha_i + \beta_i r_{m,\tau} + e_{i,\tau}, \quad (5)$$

and define abnormal return:

$$AR_{i,\tau} = r_{i,\tau} - \hat{\alpha}_i - \hat{\beta}_i r_{m,\tau}. \quad (6)$$

Cumulative abnormal return around the announcement is:

$$CAR_i(-1, 1) = \sum_{\tau=-1}^1 AR_{i,\tau}. \quad (7)$$

6.2 OLS model for announcement returns (Table 4, Panel A)

For deal d by firm i in year t :

$$CAR_d(-1, 1) = \alpha + \beta \text{China-director}_{i,t} + \gamma' W_d + \lambda' B_{i,t} + \phi' F_{i,t} + \delta_t + \psi_{\text{industry}(i)} + \varepsilon_d, \quad (8)$$

where W_d are deal-type indicators (M&A, JV, SA), $B_{i,t}$ are board controls, and $F_{i,t}$ are firm controls.

Interpretation of β . A positive β means that, controlling for deal type and firm/board attributes, firms with more outside directors with China experience create more shareholder value when announcing China-related deals.

6.3 IV model for announcement returns (Table 4, Panel B)

To address endogeneity of *China-director* in deal outcomes, the paper instruments it using the log number of Chinese immigrants in the firm's headquarters state-year:

$$\text{First stage: } \text{China-director}_{i,t} = \pi_0 + \pi_1 \log(\text{Immigrants}_{\text{state}(i),t}) + \Pi' \text{controls} + \delta_t + \psi_{\text{industry}(i)} + u_{it}, \quad (9)$$

$$\text{Second stage: } CAR_d(-1, 1) = \alpha + \beta \widehat{\text{China-director}}_{i,t} + \Gamma' \text{controls} + \delta_t + \psi_{\text{industry}(i)} + \varepsilon_d. \quad (10)$$

6.4 Operating performance after China deals (discussed in the paper)

The paper also tests whether higher announcement returns map into better post-deal fundamentals by regressing changes in industry-adjusted operating performance on *China-director*, using a pre/post window around completion (the main results are reported in the Online Appendix). A simplified

representation is:

$$\Delta ROA_i = \alpha + \beta \text{China-director}_{i,t} + \Gamma' \text{controls} + \delta_t + \psi_{\text{industry}(i)} + \varepsilon_i, \quad (11)$$

where ΔROA compares post-completion average performance to pre-announcement performance.

7 Market and labor-market valuation of China-experience directors

7.1 Appointment announcement returns (Table 5)

The paper studies a matched sample of appointment events (treatment = appointing a China-experience outside director; control = appointing a similar non-China-experience outside director). Abnormal returns are computed around the 8-K filing accepted date, and the main regression is:

$$CAR(-5, 1) = \alpha + \beta \mathbf{1}\{\text{China-experience director appointment}\} + \Gamma' \text{director controls} + \Lambda' \text{firm controls} + \varepsilon. \quad (12)$$

A positive β indicates that investors view the appointment of China-experience directors as value relevant.

7.2 Director labor market: additional board seats (Table 6)

The director-level design matches directors with China experience pre-2000 to similar directors without such experience. For each horizon $t \in \{2000, \dots, 2005\}$:

$$\Delta \text{Seats}_d(t) = \text{Seats}_d(t) - \text{Seats}_d(1999) = \alpha_t + \beta_t \mathbf{1}\{\text{director has China experience pre-2000}\} + \varepsilon_{d,t}. \quad (13)$$

The paper also considers a high-minus-low-NTR-gap version to show that new board seats are disproportionately in high-demand firms.

8 Geography and the monitoring–advisory trade-off (Table 7)

8.1 Residence decomposition

Define:

$$\text{China-director}_{i,t}^{US} = \frac{\#\{\text{China-experience outside directors domiciled in the U.S.}\}}{\#\{\text{board size}\}},$$

$$\text{China-director}_{i,t}^{Foreign} = \frac{\#\{\text{China-experience outside directors domiciled abroad}\}}{\#\{\text{board size}\}}.$$

Table 7 includes both in regressions to see which component drives advisory benefits and whether monitoring is weakened.

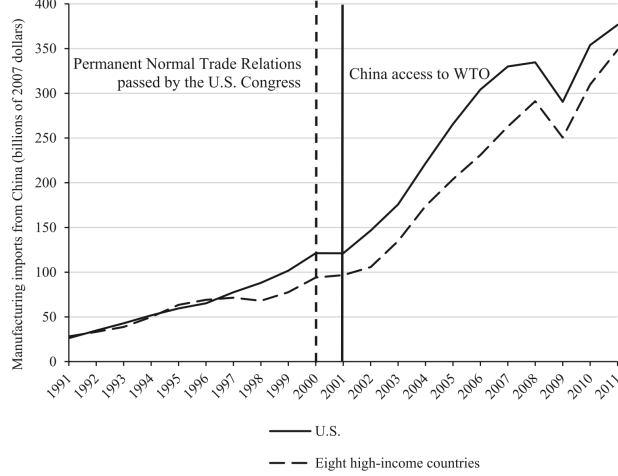


Fig. 1. Imports from China by manufacturing firms in the U.S. and eight other high-income countries. This figure depicts the value (in billions of 2007 dollars) of imported items from China from 1991 to 2011 by manufacturing firms in the U.S. (solid line) and eight high-income countries including Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland (dashed line).

8.2 Advisory outcomes

For China deals, Table 7 estimates:

$$CAR(-1, 1) = \alpha + \beta_{US} China-director^{US} + \beta_F China-director^{Foreign} + \Gamma' controls + \delta_t + \psi_{industry} + \varepsilon. \quad (14)$$

It also estimates an analogous model for post-deal operating performance.

8.3 Monitoring outcomes

To assess whether adding expert directors weakens monitoring, Table 7 examines:

- CEO incentives (CEO delta) and CEO pay, estimated with firm fixed effects,
- forced CEO turnover, financial restatement, and low meeting attendance, estimated via conditional logit with firm fixed effects.

The presence of performance interactions (industry-adjusted returns) is intended to capture whether the pay/performance relation changes with director composition.

9 Interpretation of all figures

9.1 Figure 1 (imports from China)

Figure 1 shows that U.S. imports from China rise sharply after the early 2000s and grow more strongly than in a set of other high-income countries. This motivates PNTR as a first-order shock and supports the use of non-U.S. China import penetration as an instrument for U.S. CIP.

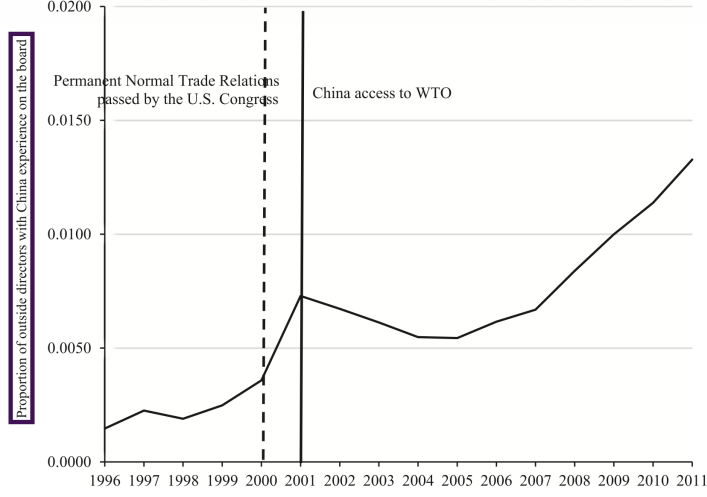


Fig. 2. Proportion of outside directors with China experience on the board. This figure depicts the mean proportion of outside directors with China experience for U.S. manufacturing firms covered in BoardEx, Capital IQ, Compustat, CRSP, EDGAR, and RiskMetrics from 1996 to 2011. An outside director is considered having China-related experience if she has worked in China or has served as a director, chief executive officer (CEO), chief financial officer (CFO), chief operating officer (COO), chief information officer (CIO), chairman, president, vice president, manager, chief officer, owner, regional CEO, regional CFO, regional COO, regional president, regional vice president, or regional manager in the Chinese division of a domestic firm or a foreign firm.

9.2 Figure 2 (time series of China-experience directors)

Figure 2 plots the mean *China-director* over time. The proportion is extremely low in the late 1990s and increases noticeably after 2000/2001, consistent with Hypothesis 1 at a descriptive level. The causal claim, however, comes from the DiD/IV regressions (Tables 2–3), not from this raw trend alone.

10 Interpretation of all tables (what each table is doing)

10.1 Table 1: Summary statistics (levels and economic context)

- **Industry exposure:** the NTR gap has substantial cross-industry dispersion (mean around 0.324; sd around 0.149).
- **Rarity of China-experience directors:** the average *China-director* is well below 1% and only a small minority of firms have at least one such director. This matters because even “small” coefficient magnitudes imply large relative changes.
- **Time trend:** the share of China-experience directors increases over time, especially after 2000, matching the PNTR narrative.

10.2 Table 2: DiD evidence for Hypothesis 1

- The coefficient on $NTR\ gap_{1999} \times Post$ is positive and statistically significant across baseline specifications.
- Results are robust to:
 - (a) using an alternative pre-period NTR gap measure,

Table 1
Summary statistics.

Panel A: Industry characteristics									
Variable	p25	Median	Mean	p75	S.D.				
NTR gap 1999	0.228	0.342	0.324	0.400	0.149				
Smoot-Hawley non-NTR 1990	0.139	0.341	0.297	0.366	0.124				
CIP in the U.S.	0.003	0.021	0.084	0.056	0.171				
CIP in eight high-income non-U.S. countries	0.003	0.018	0.064	0.058	0.112				
Panel B: Firm characteristics									
Variable	p25	Median	Mean	p75	S.D.				
China-director	0	0	0.007	0	0.036				
Firms having an outside director with China experience (indicator)	0	0	0.047	0	0.213				
Firm size (\$Millions)	58	270	2,529	1,226	8,364				
Tobin's q	1.215	1.743	2.673	2.880	3.096				
Return volatility	0.025	0.035	0.125	0.051	0.773				
Diversification (indicator)	0	0	0.397	1	0.489				
Leverage	0.001	0.115	0.196	0.276	0.331				
ROA	-0.108	0.049	-0.053	0.115	0.282				
Foreign sales ratio	0	0.199	0.274	0.479	0.290				
Board size	6	8	8	9	2.299				
Board age	57	60	60	63	4.866				
Board independence	0.600	0.750	0.705	0.833	0.170				
Board tenure	5.250	7.667	8.390	10.750	4.475				
Panel C: Director characteristics									
Year	Full sample of directors			Subsample of outside directors with China experience			b/a	Mean China-director (firm-level)	
	No. of obs. (a)	Mean tenure	Mean age	No. of obs. (b)	Mean tenure	Mean age			Mean years of China experience
1996	2,068	9.68	59.54	4	8.00	61.75	-	0.19%	0.16%
1997	2,311	9.42	59.35	6	6.67	62.17	-	0.26%	0.24%
1998	2,604	9.00	59.31	7	6.14	62.14	-	0.27%	0.20%
1999	2,703	9.07	59.42	7	6.86	63.43	-	0.26%	0.22%
2000	2,835	8.91	59.15	10	5.70	61.50	-	0.35%	0.34%
2001	4,344	8.76	58.60	32	4.94	58.13	11.95	0.74%	0.68%
2002	4,398	8.74	58.83	29	6.17	59.86	11.80	0.66%	0.59%
2003	7,547	8.57	58.59	48	5.35	58.27	12.24	0.64%	0.58%
2004	8,579	8.33	58.67	49	5.67	59.27	11.48	0.57%	0.51%
2005	8,881	8.30	58.99	51	5.10	58.57	12.38	0.57%	0.49%
2006	8,872	8.23	59.24	56	4.79	58.45	12.60	0.63%	0.53%
2007	8,783	8.14	59.44	63	4.10	56.56	12.77	0.72%	0.64%
2008	8,513	8.46	60.01	80	4.08	57.16	13.57	0.94%	0.85%
2009	8,060	8.67	60.51	88	4.26	57.82	12.96	1.09%	0.98%
2010	7,671	8.86	60.80	95	4.07	58.84	12.34	1.24%	1.09%
2011	7,754	8.69	61.02	120	3.58	58.40	11.71	1.55%	1.34%

Table 2
Effects of the passage of Permanent Normal Trade Relations (PNTR) on board composition: difference-in-differences tests.

Independent variable	Full sample period							Subsample period (year -3 to year +3)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
NTR gap 1999 × Post	0.013** (2.242)	0.012** (2.080)	0.012** (2.075)					0.012* (1.729)
NTR gap 1990 × Post				0.018*** (2.985)	0.017*** (2.696)	0.016** (2.511)		
CIP in the U.S.							0.013*** (4.819)	
Log (board size)		0.005*** (3.357)	0.005*** (3.419)		0.005*** (3.319)	0.005*** (3.381)	0.005*** (3.327)	0.007* (1.911)
Board independence		0.001 (0.507)	0.001 (0.526)		0.001 (0.530)	0.001 (0.545)	0.001 (0.670)	0.003 (0.841)
Log (firm size)		-0.001 (-1.524)	-0.001 (-1.520)		-0.001 (-1.379)	-0.001 (-1.389)	-0.001 (-1.467)	-0.001 (-0.704)
Tobin's q		-0.000 (-0.390)	-0.000 (-0.438)		-0.000 (-0.417)	-0.000 (-0.465)	-0.000 (-0.366)	0.000 (0.536)
Return volatility		0.000 (0.008)	0.000 (0.065)		0.000 (0.003)	0.000 (0.057)	-0.000 (-1.020)	0.002 (1.408)
Diversification (indicator)		-0.001 (-0.640)	-0.001 (-0.692)		-0.001 (-0.708)	-0.001 (-0.747)	-0.001 (-0.803)	0.001 (0.817)
Leverage		0.001 (0.512)	0.001 (0.604)		0.001 (0.572)	0.001 (0.658)	0.001 (0.727)	-0.003 (-0.598)
ROA		0.002** (2.161)	0.002** (2.226)		0.002** (2.110)	0.002** (2.175)	0.002** (2.184)	-0.008 (-1.334)
Foreign sales ratio		0.004** (2.066)	0.004** (2.092)		0.003* (1.938)	0.003** (1.975)	0.004** (2.232)	-0.009** (-2.299)
Revealed NTR			-0.095*** (-2.909)			-0.089*** (-2.742)		
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of observations	14,199	13,771	13,771	14,199	13,771	13,771	13,454	2,582
Adj. R^2	0.617	0.617	0.617	0.617	0.617	0.617	0.621	0.692

Table 3
Effects of the passage of Permanent Normal Trade Relations (PNTR) on board composition: instrumental variables approach.

Independent variable	Using Smoot-Hawley non-NTR 1990 as an instrumental variable				Using CIP in eight high-income non-U.S. countries as an instrumental variable	
	Without controlling for revealed NTR		Controlling for revealed NTR		CIP in the U.S.	China-director
	NTR gap 1999 × Post	China-director	NTR gap 1999 × Post	China-director		
	First stage	Second stage	First stage	Second stage		
	(1)	(2)	(3)	(4)	(5)	(6)
NTR gap 1999 × Post		0.021*** (2.843)		0.019*** (2.577)		
Smoot-Hawley non-NTR 1990 × Post	0.749*** (133.781)		0.756*** (135.254)			
CIP in the U.S.						0.027*** (7.602)
CIP in eight high-income non-U.S. countries					1.027*** (118.957)	
Log (board size)	-0.001 (-0.817)	0.005*** (3.348)	-0.002 (-1.112)	0.005*** (3.413)	-0.014*** (-3.800)	0.005*** (3.364)
Board independence	0.001 (0.321)	0.001 (0.508)	0.001 (0.247)	0.001 (0.526)	-0.006 (-1.202)	0.002 (0.692)
Log (firm size)	0.002*** (4.029)	-0.001 (-1.475)	0.002*** (4.121)	-0.001 (-1.482)	0.005*** (4.008)	-0.001 (-1.380)
Tobin's <i>q</i>	-0.000*** (-3.929)	-0.000 (-0.336)	-0.000*** (-3.728)	-0.000 (-0.396)	0.001** (2.364)	-0.000 (-0.264)
Return volatility	-0.000 (-0.160)	0.000 (0.005)	-0.000 (-0.413)	0.000 (0.063)	-0.001 (-1.216)	-0.000 (-0.923)
Diversification (indicator)	-0.001 (-0.611)	-0.001 (-0.703)	-0.000 (-0.455)	-0.001 (-0.741)	-0.003 (-1.371)	-0.001 (-0.840)
Leverage	0.005*** (4.471)	0.001 (0.480)	0.005*** (4.106)	0.001 (0.579)	-0.001 (-0.523)	0.001 (0.749)
ROA	-0.002** (-2.262)	0.002** (2.160)	-0.002*** (-2.583)	0.002** (2.226)	0.000 (0.273)	0.002** (2.133)
Foreign sales ratio	-0.005*** (-2.603)	0.004** (2.012)	-0.005*** (-2.808)	0.004** (2.051)	-0.034*** (-8.252)	0.004** (2.233)
Revealed NTR			0.407*** (12.630)	-0.095*** (-2.910)		
Cragg and Donald (1993) <i>F</i> -statistic	<i>p</i> -value < 0.001		<i>p</i> -value 0.001		<i>p</i> -value 0.001	
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of observations	13,771	13,771	13,771	13,771	13,454	13,454
Adj. <i>R</i> ²	0.977	-	0.977	-	0.920	-

- (b) controlling for *Revealed NTR* (actual duties paid / customs value),
- (c) using China import penetration (CIP) as an alternative exposure proxy,
- (d) restricting the window around PNTR (1997–2003, excluding 2000).

- Interpretation: after PNTR, firms in more-exposed industries appoint relatively more outside directors with China experience.

10.3 Table 3: IV evidence for Hypothesis 1

- **First stage:** Smoot–Hawley non-NTR tariffs strongly predict the post-PNTR NTR-gap exposure term (very strong relevance; weak-IV concerns are rejected).
- **Second stage:** instrumented exposure has a positive and statistically significant effect on *China-director*, reinforcing the interpretation that PNTR causes increased demand for China-experience directors.
- **CIP IV:** using non-U.S. China import penetration as an instrument for U.S. CIP yields a positive predicted effect on board composition, supporting robustness.

Table 4
Effects of outside directors with China experience on announcement returns for U.S. firms pursuing *China deals*.

Panel A: OLS regressions								
Independent variable	China deals							
	M&As + JVs + SAs	M&As	JVs	SAs	M&As + JVs + SAs	M&As + JVs + SAs	M&As + JVs + SAs	M&As + JVs + SAs
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
China-director	0.191** (2.189)	0.854* (1.665)	0.240* (1.745)	0.246* (1.850)	0.221** (2.459)	0.192** (2.152)	0.302* (1.701)	
First China deal (indicator)						0.000 (0.026)		
Past China investment experience (indicator)						0.002 (0.217)		
Executive China-director							−0.188 (−1.056)	
China-director with recent work experience								0.487* (1.728)
China-director with early work experience								0.103 (1.058)
Log (board size)	0.021 (1.323)	−0.022 (−0.343)	−0.057** (−2.215)	0.094* (1.903)	0.013 (0.810)	0.023 (1.472)	0.041 (1.611)	0.052* (1.802)
Board independence	−0.014 (−0.483)	0.003 (0.021)	−0.042 (−1.142)	−0.052 (−0.485)	−0.029 (−0.816)	−0.013 (−0.437)	−0.033 (−0.696)	−0.041 (−0.858)
Panel B: 2SLS regressions								
Independent variable	China-director		CAR (−1, 1)		China-director		CAR (−1, 1)	
	First stage	Second stage	First stage	Second stage	First stage	Second stage	First stage	Second stage
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
China-director			0.821* (1.646)				0.603* (1.733)	
Log (immigrants)		0.007** (2.380)				0.008*** (2.640)		
Control variables	Column 1 of Panel A	Column 1 of Panel A	Column 5 of Panel A	Column 5 of Panel A	Column 1 of Panel A	Column 1 of Panel A	Column 5 of Panel A	Column 5 of Panel A
Cragg and Donald (1993) F-statistic	p-value = 0.018	–	p-value = 0.009	–	p-value = 0.009	–	p-value = 0.009	–
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of observations	234	234	234	234	234	234	234	234
Adj. R ²	0.064	–	0.062	–	0.062	–	0.062	–

10.4 Table 4: Advisory role via China-deal announcement returns (Hypothesis 2)

- **Panel A (OLS):** higher *China-director* is associated with higher $CAR(-1, 1)$ for China deals, both in the pooled sample and when splitting by deal type.
- **Director heterogeneity:** “recent” China experience appears more valuable than “early” China experience.
- **Panel B (2SLS):** instrumenting *China-director* with Chinese-immigrant supply yields a positive second-stage effect, suggesting the advisory result is not purely selection on unobservables.

10.5 Table 5: Market reaction to appointing China-experience directors

- **Panel A (matched univariate):** treatment firms have significantly higher $CAR(-5, 1)$ around 8-K filings when appointing China-experience directors than matched control appointment events.
- **Panel B (regressions):** the appointment indicator remains positive and significant after controlling for director credentials and firm characteristics.
- Interpretation: investors view these directors as providing valuable expertise (consistent with an advisory channel).

Table 5
Announcement returns for U.S. firms that appoint outside directors with China experience.

Panel A: Univariate tests of the differences in CARs (−5, 1) between treatment and control firms				
CARs (−5, 1)	All firms	Treatment firms appointing directors with China experience (N = 39): a	Control firms appointing directors without China experience (N = 39): b	Test of difference (a − b)
Mean	0.930	2.313**	−0.453	2.766**
Median	−0.021	0.991**	−1.194	2.185**
Panel B: OLS regressions of CARs (−5, 1) for firms that appoint directors with and without China experience				
Independent variable	CAR (−5, 1)			
	(1)	(2)	(3)	
China-director appointment (indicator)	0.036*** (2.726)	0.034** (2.298)	0.034** (2.068)	
Director tenure			0.003 (0.968)	
Director age			−0.001 (−0.392)	
Director from Ivy League university (indicator)			−0.035 (−1.456)	
Director with PhD degree (indicator)			0.026 (1.372)	
Director with JD degree (indicator)			−0.010 (−0.275)	
Director with MBA degree (indicator)			0.018 (0.945)	
Director with foreign experience (indicator)			−0.002 (−0.111)	
Director with finance experience (indicator)			0.007 (0.174)	
Director with manufacturing experience (indicator)			−0.004 (−0.188)	
Director with executive experience (indicator)			−0.010 (−0.416)	
Control variables (Column 2 of Table 2)	No	Yes	Yes	
Year fixed effects	Yes	Yes	Yes	
Industry fixed effects	Yes	Yes	Yes	
No. of observations	78	76	76	
Adj. R ²	0.041	0.017	−0.065	

Table 6
Additional directorships of outside directors with China experience after the passage of Permanent Normal Trade Relations (PNTR): director-level analysis.

	t = 2000	t = 2001	t = 2002	t = 2003	t = 2004	t = 2005
Independent variable	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Change in the number of board seats from 1999 to year t						
Treatment (indicator)	0.160** (2.226)	0.182** (2.262)	0.160** (2.034)	0.297*** (3.197)	0.298*** (2.981)	0.408*** (3.740)
No. of observations	374	374	374	355	348	321
Adj. R ²	0.011	0.012	0.009	0.024	0.021	0.031
Panel B: Change in the number of board seats in high-NTR gap firms in excess of the number of board seats in low-NTR gap firms from 1999 to year t						
Treatment (indicator)	0.171** (2.222)	0.203* (1.736)	0.171 (1.337)	0.337* (1.904)	0.458** (2.548)	0.562*** (3.070)
No. of observations	374	374	374	355	348	321
Adj. R ²	0.011	0.006	0.002	0.007	0.015	0.020

10.6 Table 6: Director labor-market rewards after PNTR

- Directors with China experience gain more board seats than matched controls immediately after PNTR, and the gap grows over 2000–2005.
- The additional seats are disproportionately in high-NTR-gap firms, consistent with a demand-shock story (firms most exposed to China need the expertise most).

10.7 Table 7: Residence and the monitoring–advisory trade-off

- **Advisory outcomes (Columns 1–2):** the positive relation with $CAR(−1, 1)$ and post-deal operating performance is driven by China-experience directors domiciled in the U.S.; foreign-domiciled directors show little evidence of value creation in these outcomes.
- **Monitoring outcomes (Columns 3–7):** U.S.-domiciled China-experience directors are

Table 7
Residence locations of outside directors with China experience.

Independent variable	Change in		Log (1 + CEO total compensation)		Forced CEO turnover (indicator)	Financial restatement (indicator)	Meeting attendance less than 75% (indicator)
	CAR (−1, 1)	ROA	CEO delta				
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	Conditional logit (5)	Conditional logit (6)	Conditional logit (7)
China-director domiciled in the U.S.: a	0.275* (1.775)	0.333** (2.208)	−0.545 (−0.719)	−0.718 (−1.292)	−12.342 (−1.094)	1.511 (0.292)	−0.845 (−0.884)
China-director domiciled in foreign countries: b	0.003 (0.020)	0.085 (0.271)	−1.314 (−1.400)	−0.798 (−0.949)	−9.446 (−0.791)	33.254* (1.733)	2.553*** (5.721)
Industry-adjusted return: c				0.103*** (4.071)	−0.816** (−2.128)		
a × c				−0.027 (−0.075)	12.469 (0.841)		
b × c				0.036 (0.066)	4.912 (0.698)		

not associated with statistically significant deteriorations in monitoring proxies. In contrast, foreign-domiciled China-experience directors are associated with poorer meeting attendance and (in one specification) higher likelihood of restatements, consistent with distance/engagement constraints.

11 Bottom line

The paper provides rare evidence that a policy-induced change in firms' environment shifts board composition toward specific, relevant expertise, and that such expertise has measurable value in corporate decisions where it should matter. The designs combine DiD and multiple IV strategies, and the results are supported by (i) market reactions to director appointments and (ii) director labor-market outcomes.

12 Conclusion and intuition

12.1 Coherent summary: what the paper concludes

This paper studies how an *exogenous demand shock for a specific director expertise* reshapes board composition and affects firm decisions where that expertise should matter. The shock is the U.S. Congress's grant of *Permanent Normal Trade Relations (PNTR)* status to China in 2000, which substantially reduced policy uncertainty around U.S.–China trade and increased U.S. firms' incentives to engage in China-related business.

The paper's main conclusions are four tightly connected results:

1. **Boards endogenously adjust their expertise mix after PNTR.** After 2000, U.S. firms appoint more outside directors with *China-related experience*, and this increase is stronger for firms in industries with higher exposure to PNTR (measured by the pre-PNTR *NTR gap*).
2. **China-experience outside directors add value in settings where their expertise is directly relevant.** Firms with a higher fraction of outside directors with China experience realize *higher announcement returns* around China-related investments (M&As, joint ventures, strategic alliances) and exhibit *better post-deal operating performance*. These patterns support an *advisory-channel* interpretation rather than a pure selection story.

3. **Financial markets and director labor markets price this expertise.** The stock market reacts more positively to the appointment of a China-experience director, and directors with China experience obtain more board seats after PNTR, especially at firms where demand for such expertise is high.
4. **Geography matters: the value-enhancing effect is driven mainly by local (U.S.-domiciled) China-experience directors.** The advisory benefits are concentrated among China-experience directors who reside in the U.S. and are therefore geographically closer to the firm. In contrast, foreign-domiciled China-experience directors are associated with weaker engagement (e.g., lower meeting attendance) and are linked in some specifications to weaker monitoring outcomes (e.g., higher likelihood of certain restatement indicators).

12.2 Economic intuition: why PNTR changes board structure and why expertise creates value

The paper’s logic can be summarized as *demand for specialized knowledge + frictions in cross-border business + endogenous board design*.

Step 1: PNTR increases the expected payoff to China-related corporate activity. By reducing the risk of large tariff increases (and therefore reducing a major source of uncertainty), PNTR raises the net expected returns to sourcing, investing, and partnering in China.

Step 2: China-related activity is expertise intensive. Cross-border decisions involve nontrivial informational and contracting frictions: differences in regulatory environments, business norms, counterparties, and operational integration. These frictions create a role for advisors who can:

- identify suitable targets/partners,
- reduce information asymmetry,
- facilitate post-deal integration and resource coordination.

Step 3: outside directors supply the missing expertise. If boards provide not only monitoring but also advising, then firms will rationally change director composition when the marginal value of a particular expertise increases. PNTR is the shock that makes China expertise more valuable, so boards tilt toward directors with China experience.

Step 4: why local directors are the effective channel. Advising (and monitoring) requires time, interaction, and ongoing involvement. Directors with relevant expertise who are geographically closer can be engaged in board processes more consistently. Foreign-domiciled directors may possess expertise but face practical frictions (distance, time zones, travel, competing obligations), which can reduce meeting attendance and weaken monitoring, attenuating the net value of their expertise.

12.3 Final takeaway

A policy-driven increase in firms' China exposure (PNTR) induces firms to re-optimize board expertise toward China-experience outside directors, and that expertise is priced by markets and improves China-deal outcomes—especially when the expert directors are locally engaged rather than geographically distant.